

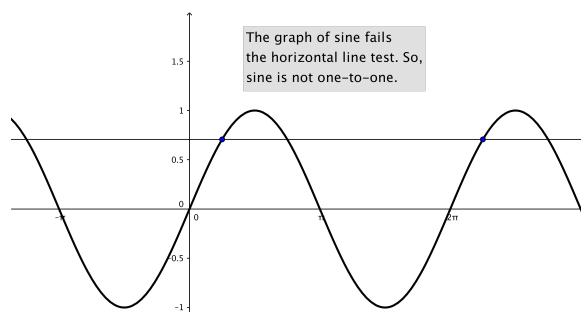
Problem 8

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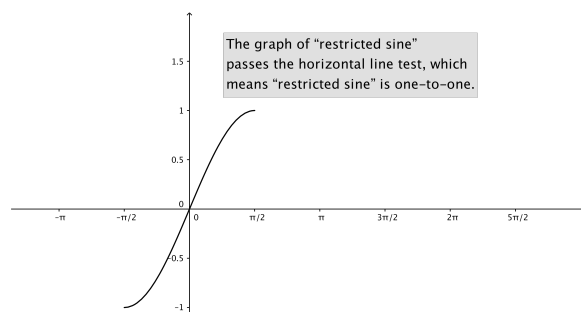
Problem 1 True or False: $\sin^{-1}(0) = \pi$.

Explanation. This statement is **false**: the correct statement is $\sin^{-1}(0) = 0$. (Why?)

Spoiler Alert: the sine function is *not* invertible since its graph fails the horizontal line test.



To produce the inverse sine we first restrict the domain of sine, to the interval $[-\pi/2, \pi/2]$, to produce an invertible function:



So, the range of $\sin^{-1}$ is $[-\pi/2, \pi/2]$. Since π is not in this range, π , is *never* a possible output of $\sin^{-1}$.